

Salvage Proton Beam Therapy in Local Recurrent Uveal Melanoma

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- **PURPOSE:** To evaluate survival and ocular outcome in recurrent uveal melanoma treated with proton beam therapy as salvage therapy.
- **DESIGN:** Retrospective, interventional case series.
- **METHODS:** We evaluated 48 patients with local recurrence of uveal melanoma after primary treatment with brachytherapy, transpupillary thermotherapy, proton beam therapy, laser photocoagulation, CyberKnife radiation, or photodynamic therapy. All patients received proton beam therapy as a salvage therapy at the Helmholtz Zentrum Berlin between July 2000 and December 2010. Kaplan-Meier analysis was used to obtain survival rates.
- **RESULTS:** The Kaplan-Meier estimator for local tumor control was 92.1% at 10 years after secondary treatment with proton beam therapy. Local recurrence developed in 3 patients; 1 of them underwent enucleation. During follow-up, 20.8% of the patients died (16.7% of metastasis, 4.1% of other causes or not specified). The most frequent surgical interventions were phacoemulsification (20.8%) and pars plana vitrectomy (10.4%). The Kaplan-Meier estimators were 77.4% for survival and 70.1% for the absence of metastasis 10 years after the primary treatment.
- **CONCLUSIONS:** Proton beam therapy as a salvage treatment resulted in high local tumor control rates in recurrent uveal melanoma, especially if the primary therapy was transpupillary thermotherapy or plaque brachytherapy. Preservation of the globe was possible in most patients. Enucleations were indicated only in case of re-recurrences of uveal melanoma, but not because of secondary complications like intractable pain or secondary glaucoma. Retreatment was associated with vision deterioration, but loss of vision remained exceptional. Further larger prospective studies are needed to confirm the presented results of our retrospective analysis. (*Am J Ophthalmol* 2014;158:948–956. © 2014 by Elsevier Inc. All rights reserved.)

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UVEAL MELANOMA IS THE MOST COMMON PRIMARY malignant ocular tumor in adults.¹ Beyond the means of enucleation, different treatment methods for local tumor control have been developed during recent decades with the goal of preservation of the globe and its function. Today, the most accepted therapies are based on radiation, including brachytherapy with episcleral plaques (eg, ¹²⁵I or ¹⁰⁶Ru), teletherapy with heavy charged particles (proton beam therapy, helium ion therapy), and CyberKnife radiotherapy. In the end, all treatment methods bear the risk of local recurrence.

Depending on the technique, tumor size, and location, recurrence rates range between 3.2% to 22% in large studies.^{2,3} The type of recurrence usually is not specified further in most studies. Recurrence may take the form of marginal recurrence, diffuse recurrence, distant recurrence, or extrascleral extension.⁴ Large tumor diameter, small tumor volume, iris root involvement, and safety margins inferior to 1 mm have been identified as risk factors for local recurrence after proton beam therapy.⁴ Patients with tumors abutting or overhanging the optic disc and those requiring more than 3 sessions for tumor control had a much higher risk for recurrence after primary transpupillary thermotherapy (TTT).³ A specific issue of plaque therapy is the treatment of peripapillary tumors because the optic nerve is a mechanical barrier. Notched plaques are a possibility for juxtapapillary melanomas, but the correct placement for postequatorial plaques is challenging and can result in malpositioning or tilting.⁵ To spare critical structures, eccentric plaque placement has been practiced in some centers, which can result in higher recurrence rates.⁶

Currently, enucleation is the most frequently used treatment for recurrent melanoma. This may be a consequence of the unclear prognostic impact of local recurrence. Several studies suggest that local recurrence is associated with a reduced outcome regarding the development of metastases and survival.^{7,8} As there is no consent for retreatment, the way of retreatment remains a single case decision. The treatment approach depends on the extent and the location of the recurrence.^{3,9} Only a few studies have analyzed the salvage treatment of recurrent choroidal melanoma with proton beam therapy.^{9,10} The goal of our study was to evaluate the survival rate and functional outcome of patients with local recurrence of choroidal melanoma retreated with proton beam therapy.